

1. Write a program using Kotlin to implement control structures and loops.

```
fun main() {  
    // Control structure (if-else)  
    val number = 7  
    if (number > 0) {  
        println("Positive number")  
    } else {  
        println("Negative number")  
    }  
  
    // For loop  
    println("Numbers from 1 to 5:")  
    for (i in 1..5) {  
        println(i)  
    }  
  
    // While loop  
    var count = 1  
    println("While loop output:")  
    while (count <= 3) {  
        println(count)  
        count++  
    }  
}
```

2. Write a program to implement object-oriented concepts in Kotlin.

```
open class Person(var name: String) {  
    open fun show() {  
        println("Name: $name")  
    }  
}  
  
class Student(name: String, var rollNo: Int) : Person(name) {  
    override fun show() {  
        println("Name: $name")  
        println("Roll No: $rollNo")  
    }  
}
```

```
fun main() {  
    val s = Student("Ravi", 101)  
    s.show()  
}
```

3. Write a program for implementation of Kotlin polymorphism.

Kotlin Program: Polymorphism

// Base class

```
open class Animal {  
    open fun sound() {  
        println("This animal makes a sound")  
    }  
}
```

// Derived classes

```
class Dog : Animal() {  
    override fun sound() {  
        println("Dog says: Woof Woof")  
    }  
}
```

```
class Cat : Animal() {  
    override fun sound() {  
        println("Cat says: Meow Meow")  
    }  
}
```

```
fun main() {  
    val myAnimal1: Animal = Dog() // Animal reference, Dog object  
    val myAnimal2: Animal = Cat() // Animal reference, Cat object  
  
    myAnimal1.sound() // Calls Dog's sound()  
    myAnimal2.sound() // Calls Cat's sound()  
}
```

4. Write a program for implementation of Kotlin inheritance.

// Parent class

```

open class Animal {
    fun eat() {
        println("Animal is eating")
    }
}

// Child class
class Dog : Animal() {
    fun bark() {
        println("Dog is barking")
    }
}

fun main() {
    val d = Dog()
    d.eat() // inherited function
    d.bark() // child function
}

```

Output:

```

Animal is eating
Dog is barking

```

5. Write a program to implement Kotlin interfaces.

```

// Interface
interface Shape {
    fun draw()
}

// Class implementing interface
class Circle : Shape {
    override fun draw() {
        println("-")
    }
}

// Main function
fun main() {
    val c = Circle()
    c.draw()
}

```

6. To create a simple Android application that displays the text “Hello Android” on the screen.

Steps

1. Open **Android Studio** → **New Project** → **Empty Activity**
2. Name: **HelloAndroid** → Language: **Kotlin** → Finish
3. Open **activity_main.xml**
4. Replace content with a **TextView**:

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/tvHello"
        android:text="Hello Android"
        android:textSize="30sp"
        android:textColor="#FF5722"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

5. Run the app → You will see **Hello Android** in the center.
7. To design a calculator app that takes two numbers as input and displays their sum when the user clicks a button.

Steps

1. Create project: **SimpleCalculator** → Kotlin → Empty Activity
2. Open **activity_main.xml** → Add 2 EditText, 1 Button, 1 TextView:

```
<LinearLayout
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp">

    <EditText
        android:id="@+id/num1"
        android:hint="Enter First Number"
        android:inputType="number"
        android:layout_width="match_parent"
```

```

        android:layout_height="wrap_content"/>

<EditText
    android:id="@+id/num2"
    android:hint="Enter Second Number"
    android:inputType="number"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<Button
    android:id="@+id/btnAdd"
    android:text="ADD"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"/>

<TextView
    android:id="@+id/tvResult"
    android:text="Result will appear here"
    android:textSize="20sp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"/>
</LinearLayout>

```

3. Open **MainActivity.kt** → Add:

```

btnAdd.setOnClickListener {
    val n1 = num1.text.toString().toInt()
    val n2 = num2.text.toString().toInt()
    val sum = n1 + n2
    tvResult.text = "Result = $sum"
}

```

4. Run → Enter numbers → Press **ADD** → Result appears.